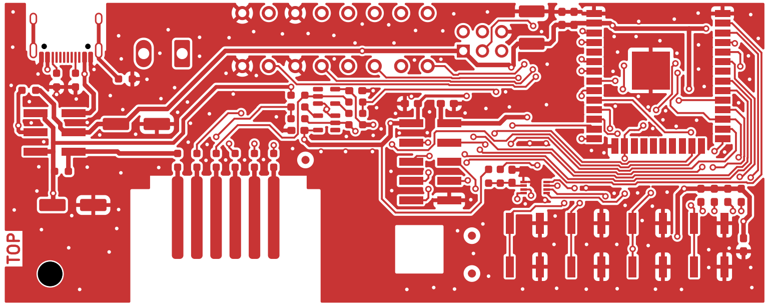
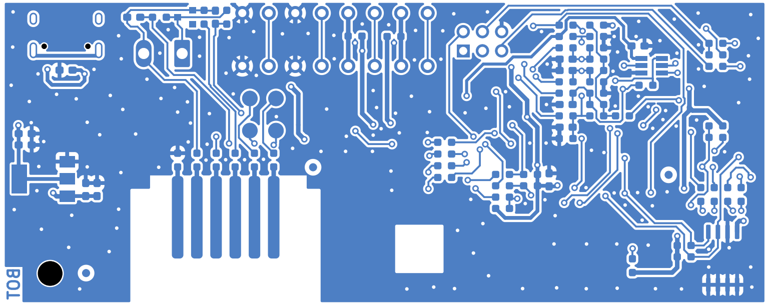
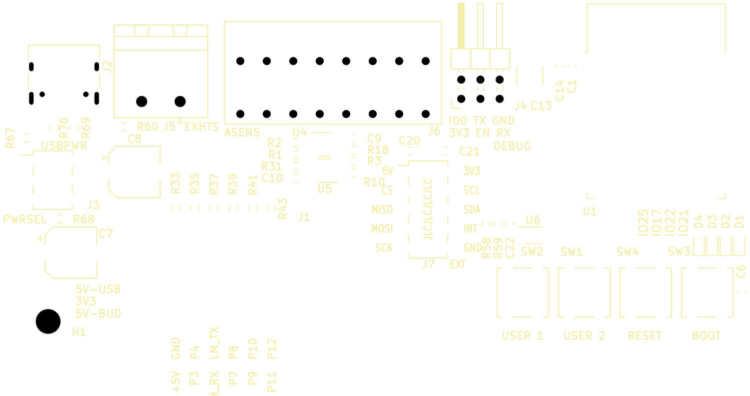
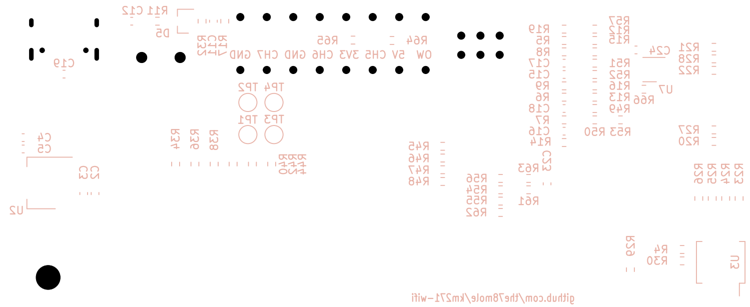
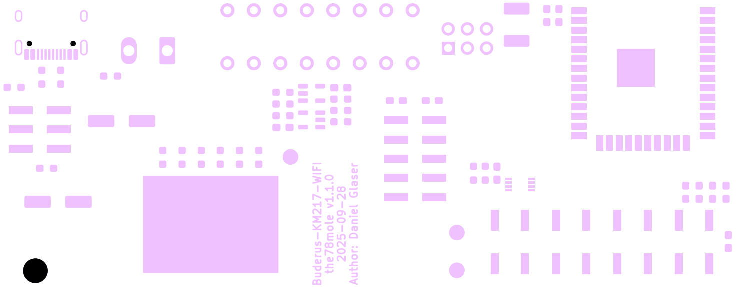

TOP

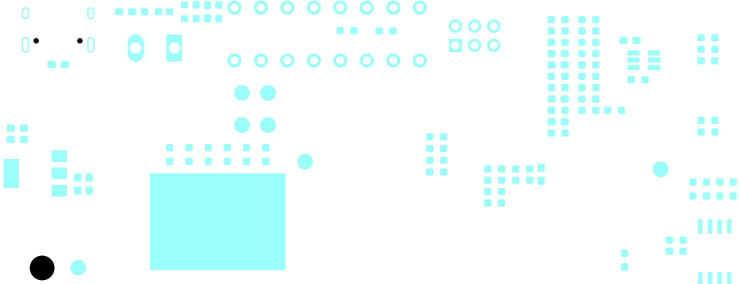


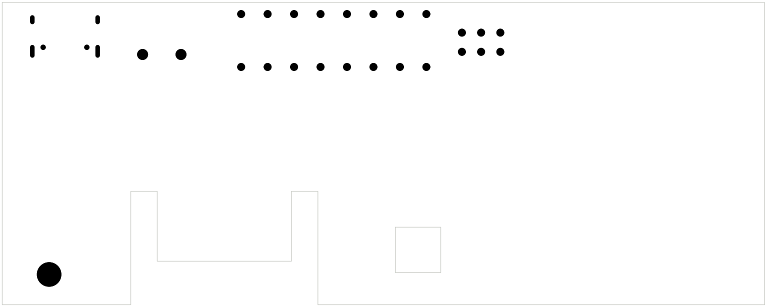
BOT

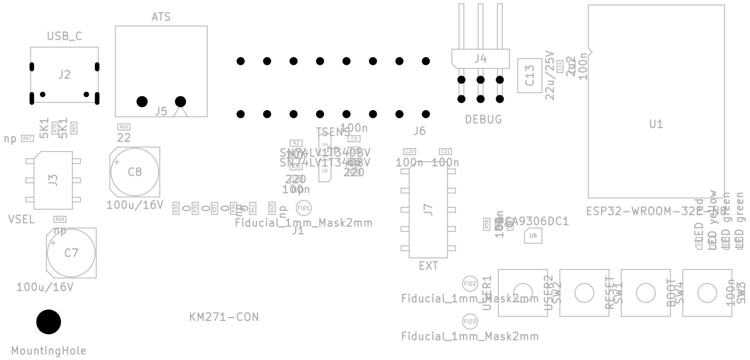

J2
R67
R70
R69
USBPWR
R60
J5
EXHTS
C8
ASENS
U4
J6
IO0
TX
GND
3V3
EN
RX
DEBUG
J4
C13
C14
C1
R2
R1
R31
C10
R18
R3
R10
C9
C20
C21
5V
CS
MISO
MOSI
SCK
3V3
SCL
SDA
INT
GND
JLCJLCJLCJLC
J7
EXT
U5
R33
R35
R37
R39
R41
R43
J1
J3
PWRSEL
R68
C7
5V-USB
3V3
5V-BUD
H1
R58
R59
C22
U6
SW2
SW1
SW4
SW3
U1
IO25
IO17
IO22
IO21
D4
D3
D2
D1
C6
USER 1
USER 2
RESET
BOOT
+5V
GND
P3
P4
LM_RX
LM_TX
P7
P8
P9
P10
P11
P12

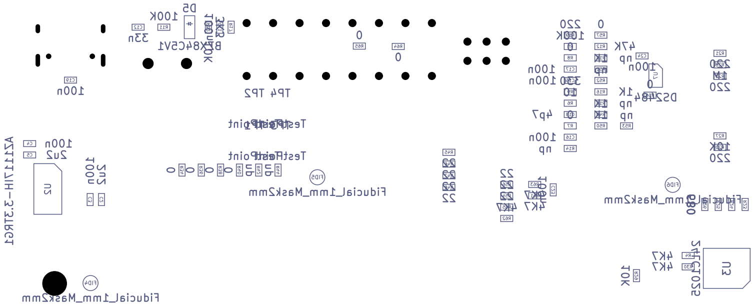

OW 5V CH5 3V3 CH6 GND CH7 GND
TP1 TP2 TP3 TP4
D5
U2
U3
U7
C24
github.com/the78mole/km271-wifi

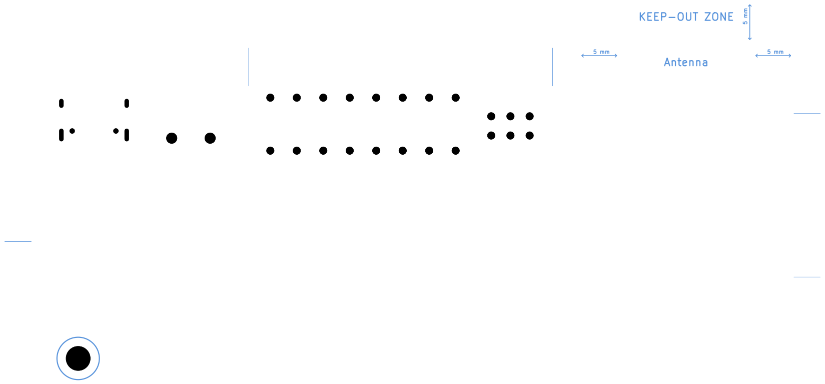

Buderus-KM217-WIFI
the78mole v1.1.0
2025-09-28
Author: Daniel Glaser






USB_C
J2
ATS
J5
J4
C13
22u/25V
2u2
100n
U1
DEBUG
J6
5K1
5K1
np
22
TSENS
100n
SN74LV1T34DBV
SN74LV1T34DBV
220
220
100n
100n
100n
J3
C8
100u/16V
VSEL
np
J7
Fiducial_1mm_Mask2mm
J1
ESP32-WROOM-32E-N8
PCA9306DC1
100n
EXT
LED red
LED yellow
LED green
LED green
C7
100u/16V
USER1
USER2
SW2
RESET
SW1
BOOT
SW4
100n
SW3
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
KM271-CON
MountingHole


Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
TestPoint
TestPoint
TP4 TP2
D5
BZX84C5V1
DS2484
AZ1117IH-3.3TRG1
U2
U3
100n
2u2
100K
33n
4p7
220
1M
10K
4K7
22
0



FAB NOTE J1 (KM271 card edge): 12 gold fingers F+B, 1.6 mm FR4 (+-10%).
Order option: Gold Fingers = YES (ENIG / hard gold on fingers), mask open over fingers.
Bevel: NOT POSSIBLE - finger edge (y=124.25) is recessed 5.75 mm between the board legs
(y=130) and JLC chamfers only boards >= 50x50 mm. Original module is used unbevelled.
Fingers end 0.2 mm before the board edge; no vias/pads within 1 mm (rule area GOLD_FINGERS).
FAB NOTES Buderus-KM217-WIFI rev v1.1.0
Stackup: 2 layers, FR4 TG135, 1.6 mm +-10 pct, outer copper 1 oz (35 um)
L1 F.Cu signal/GND pour | core 1.5 mm | L2 B.Cu signal/GND pour
Finish: ENIG (gold fingers). Gold fingers: YES, bevel: NO (see J1 note)
Mask green, silk white, vias tented. Min track/space 0.20/0.20 mm, min via 0.3/0.6 mm
Assembly: Standard PCBA, both sides. BOM/CPL: production/bom.csv, production/positions.csv